

Analyzing a Game Show

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0.1 Intro

You're playing a game show with a straightforward premise. You start with \$1 and flip 100 coins, following this algorithm:

1. Set $d_0 = 1$, your initial dollar amount.
2. Pick a constant $k \in [0, 1]$: that is, $0 \leq k \leq 1$.
3. For $i = 1, 2, \dots, 100$, flip a fair coin and define

$$d_i = \begin{cases} (1 + k) \cdot d_{i-1} & \text{if the coin lands heads} \\ (1 - k) \cdot d_{i-1} & \text{if the coin lands tails.} \end{cases}$$

That is, you gain k times your current dollar amount if the coin is heads, and you lose the same amount if not.

Your goal is to end the game with the highest dollar amount you can. In other words, you aim to maximize d_{100} . Your only input during the game is the choice of the number k in step (2), leading us to the first question of this note:

Question 0.1.1. What value of k maximizes $E(d_{100})$, the expected value of d_{100} ?

Our team found multiple ways to the answer of Question 0.1.1. One way is by defining, for each i , the random variable

$$X_i = \frac{d_i}{d_{i-1}} = \begin{cases} 1 + k & \text{if the } i\text{-th coin is heads} \\ 1 - k & \text{if the } i\text{-th coin is tails.} \end{cases}$$

What's the probability distribution of X_i ? Since we are flipping fair coins, X_i has a 50% chance of being $1 + k$, and a 50% chance of being $1 - k$. In particular, the expected value of X_i is

$$\begin{aligned} E(X_i) &= \left(P(X_i = 1 + k) \cdot (1 + k) \right) + \left(P(X_i = 1 - k) \cdot (1 - k) \right) \\ &= 0.5 \cdot (1 + k) + 0.5 \cdot (1 - k) \\ &= 1 \end{aligned}$$

The next thing to notice is that the final dollar amount d_{100} is just the product of all these X_i together:

$$d_{100} = X_1 \cdot X_2 \cdot \dots \cdot X_{99} \cdot X_{100} = \prod_{i=1}^{100} X_i$$

Since the X_i are independent of each other, we can now compute $E(d_{100})$ by multiplying the expectations of each X_i :

$$E(d_{100}) = E\left(\prod_{i=1}^{100} X_i\right) = \prod_{i=1}^{100} E(X_i) = 1. \quad (1)$$

Thus we arrive at the answer to Question 0.1.1. In truth, we've realized two things:

1. The choice of k does NOT affect the player's expected earnings.
2. The player's money is expected to be the same at the end of the game as it was at the beginning.

That's kind of interesting, but it's also kind of dull, too. Everyone on the team immediately started coming up with variations of the game in order to see how the player's strategy might evolve. For example, several people asked the following question.

Question 0.1.2. Suppose the coin is not necessarily fair, but instead has a probability of heads equal to h . What value of k should the player choose to maximize $E(d_{100})$ based on h ?

Our analysis proceeds in much the same way as before. In this case, each X_i now has expectation

$$\begin{aligned} E(X_i) &= h \cdot (1 + k) + (1 - h) \cdot (1 - k) \\ &= h + hk + 1 - h - k + hk \\ &= 1 - 2hk - k \\ &= 1 + k(2h - 1). \end{aligned}$$

By the same logic as before, we then have

$$E(d_{100}) = \prod_{i=1}^{100} E(X_i) = (1 + k(2h - 1))^{100}. \quad (2)$$

Fantastic! Now the dependence on the player's choice k is in the formula for their expected payout. We can also see that when the coin is fair, i.e. when $h = 1/2$, the contribution from k vanishes.

In general, though, how can the player maximize their expectation using this formula? There are three cases, summarized in Table 1.

Table 1: Game possibilities

Case	Value of h	Optimal k	$E(d_{100})$
Biased against player	$h < 1/2$	$k = 0$	1
Fair coin	$h = 1/2$	Any k	1
Biased for player	$h > 1/2$	$k = 1$	$(2h)^{100}$

On paper, it looks like the case when $h > 1/2$ is best for the player. After all, the player's expected payout is the highest in this case: if $h > 1/2$ then $2h > 1$, and so $(2h)^{100} > 1$. Moreover, the value of this payout skyrockets extremely quickly as h increases above 50%.

Here, however, is where we see the failure of focusing only on the expected value. Choosing $k = 1$ means the player wagers all of their money on every coin flip, so the only way they finish the game with any money at all is if every single coin flip be heads:

Table 2: Probability distribution for $h > 1/2$ and $k = 1$

Number of heads	Probability	Payout
100	h^{100}	$(2h)^{100}$
99 or fewer	$1 - h^{100}$	0

That is a bimodal distribution *par excellence*. To help understand how unlikely it is for the player to finish the game without losing all their money, note that in order for the **median** payout to not be 0, we require the probability of flipping heads to satisfy

$$h > \sqrt[100]{0.5} = 0.993\dots$$

Suffice it to say that many people would not call this strategy “optimal.”

At this point, it's natural to wonder what the distribution of d_{100} looks like in general. What are the possible values of d_{100} ? Remembering that d_{100} is the product of the X_i 's, we can start listing out these possible values:

1. If no coin lands heads, then $X_i = 1 - k$ for each i , and so $d_{100} = (1 - k)^{100}$.

2. If exactly one coin lands heads, then $d_{100} = (1+k)(1-k)^{99}$.
3. In general, if exactly i coins land heads, then $d_{100} = (1+k)^i(1-k)^{100-i}$.

Moreover, the number of ways exactly i coins land heads is the binomial coefficient

$$\binom{100}{i} = \frac{100!}{i!(100-i)!}.$$

We obtain the following formula for the probability distribution of d_{100} : for each $i = 0, 1, \dots, 100$,

$$P(d_{100} = (1+k)^i(1-k)^{100-i}) = \binom{100}{i} \cdot h^i \cdot (1-h)^{100-i} \quad (3)$$

Several members of the team pointed out that this looks quite a bit like a binomial distribution. For comparison, the distribution of a binomial random variable B with 100 trials and probability of success h is

$$P(B = i) = \binom{100}{i} \cdot h^i \cdot (1-h)^{100-i}$$

In other words, the *probability values* (y -values in the histogram) of the distribution of d_{100} match with a binomial distribution, but the *outcome values* (x -values) are different. To bridge the gap between these distributions, we can make a few observations. For this part, we'll have to require that $k \neq 1$:

1. The minimum value of B is 0, and each subsequent value is just one more than the previous.

$$0 \xrightarrow{+1} 1 \xrightarrow{+1} \dots \xrightarrow{+1} 100$$

2. The minimum value of d_{100} is $(1-k)^{100}$, and each subsequent value is $\frac{1+k}{1-k}$ times more than the previous one (this is why we can't have $k = 1$):

$$(1-k)^{100} \xrightarrow{\times \frac{1+k}{1-k}} (1+k)(1-k)^{99} \xrightarrow{\times \frac{1+k}{1-k}} \dots \xrightarrow{\times \frac{1+k}{1-k}} (1+k)^{100}$$

Therefore, if we set

$$\alpha = \frac{1+k}{1-k} \quad \text{and} \quad \beta = \log_{\alpha}((1-k)^{100}),$$

then a little algebra will show that

$$B = \log_{\alpha}(d_{100}) - \beta.$$

In other words, d_{100} has a *log-binomial*. In particular, any analysis of d_{100} can usually be done by studying the binomial variable B and then translating the results appropriately.